Should the Fusion Weight Change with the Question?

A Study of Continually Learned Fusion Weights for Keyframe Search

Jianxiang He

CV805 Course Project

Mohamed bin Zayed University of Artificial Intelligence

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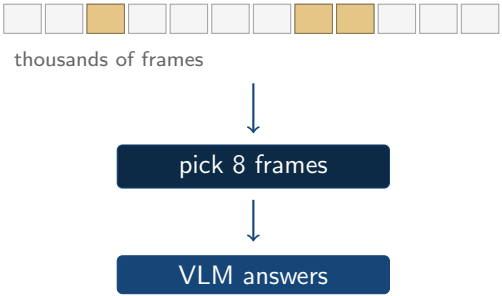


Outline

- Problem
- Setup
- Q1: Does the Best Weight Depend on the Question?
- Q2 & Q3: What Does Continual Learning Learn?
- Q4: Do Better Frames Give Better Answers?
- Lessons and Conclusion

Long videos, few frames

- A video can last tens of minutes.
- A vision-language model (VLM) only takes **about 8 frames**.
- If the 8 frames miss the moment the question is about, the model cannot answer.



Keyframe search

Before calling the VLM, find the frames that are most related to the question.

VSI: two clues, one weight

For every second of the video, VSI computes two scores:

- **Visual**: do the question's objects appear?
- **Subtitle**: does the subtitle match the question?

Then it mixes them and keeps the top 8 seconds:

$$S(t) = (1 - \lambda)\,\text{visual}(t) + \lambda\,\text{subtitle}(t)$$

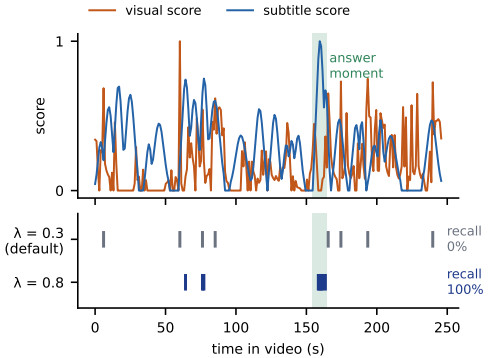
$\lambda = 0$: picture only, $\lambda = 1$: subtitles only.
VSI uses $\lambda = 0.3$ **for every question**.

Our question

Is one fixed λ right for all questions?

"What does the subtitle say when. . . " should trust subtitles.
"What color is the man's shirt" should trust the picture.

One real example



Question: "...when the spoon places the meat onto the mashed potatoes and the subtitle mentions 'Just pour the...'".

With the default $\lambda = 0.3$ all 8 frames **miss** the answer moment; with $\lambda = 0.8$ they **hit** it.

Our idea: learn the weight

- Train a **small predictor** that looks at a question and outputs its own λ.
- New question types keep appearing, so the predictor learns them **one after another**.
- It should learn new types **without forgetting** old ones.
- This setting is called **continual learning**.

Four questions we ask

- Q1 Does the best λ depend on the question type?
- Q2 Can continual learning learn it? Does it forget?
- Q3 If not, what is holding it back?
- Q4 Do better frames give better answers?

Experimental setup

Data: LongVideoBench, 753 questions (one per video)

Four question types (tasks), learned in order:

- T1 text perception (127) T2 visual perception (237)
- T3 text relation (171) T4 visual relation (218)

70% train / 30% test; 3 task orders $\times$ 3 seeds = **9 runs**

Predictor: small 2-layer network, 54 inputs computed from the two score curves

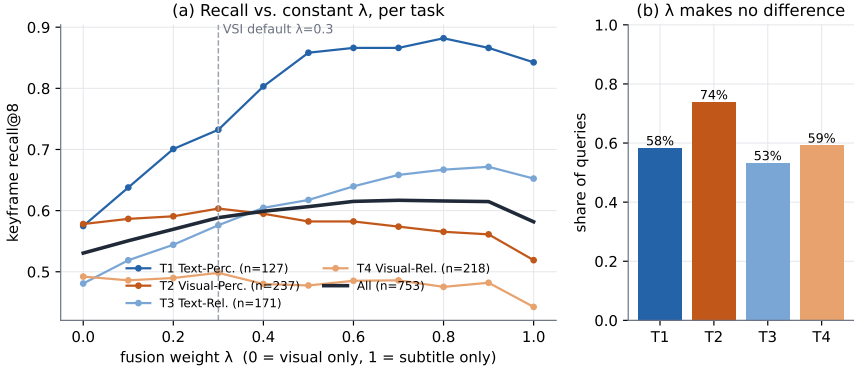
Continual learning methods: Naive, EWC, LwF, ER (replay), kNN, and Joint (trains on everything, as a reference)

Baselines:

- default $\lambda = 0.3$
- **one λ tuned on the training set**
- one λ per question type
- oracle: best λ for every question (upper limit)

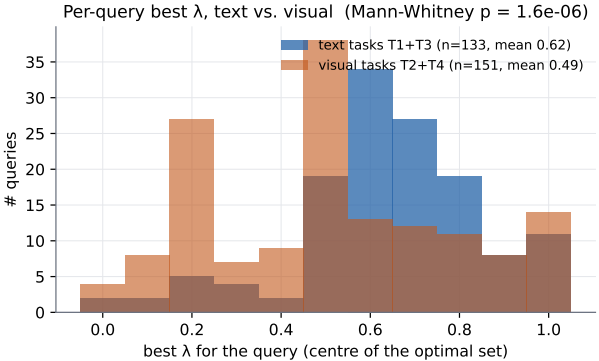
Metrics: recall@8 = share of answer moments hit by the 8 frames; ACC = final average; FM = forgetting

Q1: types want different weights



Text tasks (blue) peak at $\lambda = 0.8$–0.9; visual tasks (orange) peak at $\lambda \leq 0.3$.
T1: moving λ from 0.3 to 0.8 raises recall from **0.73 to 0.88**. But **53–74%** of questions do not react to λ at all.

Q1: question by question



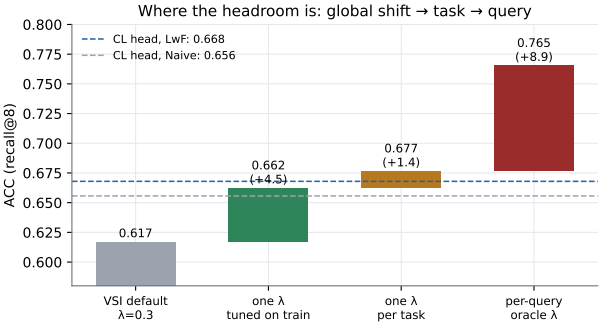
Only the 284 questions where λ matters:

- text questions: mean best $\lambda = \mathbf{0.62}$
- visual questions: mean best $\lambda = \mathbf{0.49}$
- $p = 1.6 \times 10^{-6}$

Answer to Q1

Yes. The best weight clearly depends on the question type.

Q1: where is the room to improve?



- Tune one global λ: **+4.5**
- One λ per type: only **+1.4** more
- Best λ per question: **+8.9** more

Takeaway

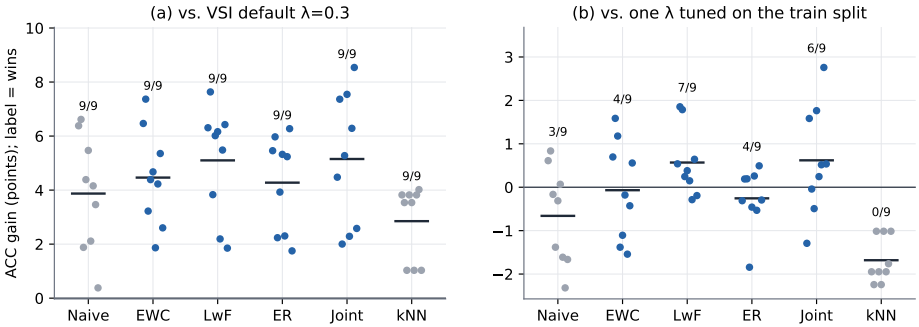
Most of the gain is in **per-question** differences, and only a minority of questions carry that signal.

Q2: main results (9 runs)

Method	ACC ↑	FM ↓	vs. tuned
Default $\lambda = 0.3$	0.617	–	−4.5
One tuned λ	0.662	–	–
One λ per type	0.677	–	+1.4
Oracle	0.765	–	+10.3
Naive	0.656	0.029	−0.7
EWC	0.662	0.023	−0.1
LwF	**0.668**	**0.015**	+0.6
ER	0.660	0.021	−0.3
Joint (ref.)	**0.668**	**0.015**	+0.6
kNN	0.645	0.005	−1.7

- All methods beat the default by **4–5 points**, in **9 of 9** runs.
- But compared with **one tuned λ**, the best methods gain only **+0.6**.
- None reaches one λ per type.
- LwF and Joint forget about half as much as Naive, but only in 6 of 9 runs.

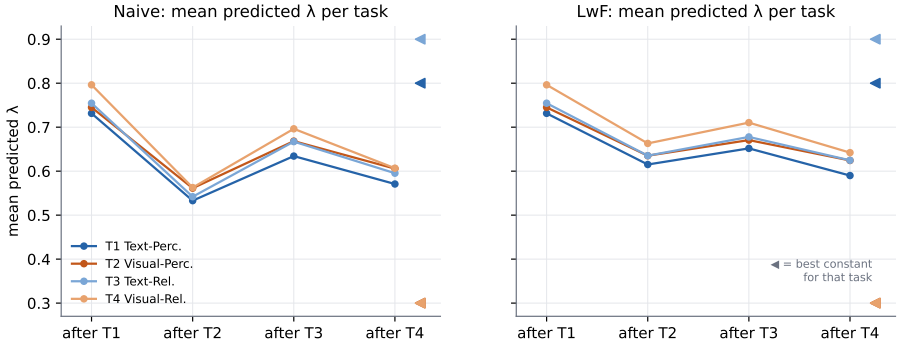
Q2: run by run



Each dot is one run. (a) Against the default: always better. (b) Against one tuned λ: **about even**.

Lesson: compare learned weights with a **tuned** constant, not with the paper's default.

Q3: one weight for every type



Average predicted λ for each question type after each training stage. Triangles = best λ for that type.

The four lines overlap at about 0.6: it learned a **better global value**, not **different values per type**.

Q3: why no differences?

1. Sparse training signal

62% of questions give the same result for every λ, so almost all learning comes from the remaining third.

2. No question-type information

The inputs only describe the two score curves, which look similar across question types. Adding the question text made it *worse* (overfitting on 753 questions).

Why forgetting is small

All types end up with similar weights, so learning a new type overwrites little.

Still visible: after T2, Naive drops all weights to 0.55; LwF stays steadier.

Q4: answering with Qwen3-VL-8B

Frames chosen by	QA accuracy
Even spacing	0.615
$\lambda = 0$ (picture only)	0.604
$\lambda = 0.3$ (default)	0.631
$\lambda = 0.7$ (best recall)	0.627
$\lambda = 1$ (subtitles only)	0.615
Oracle λ	0.649
LwF predictor*	+0.4 vs. default

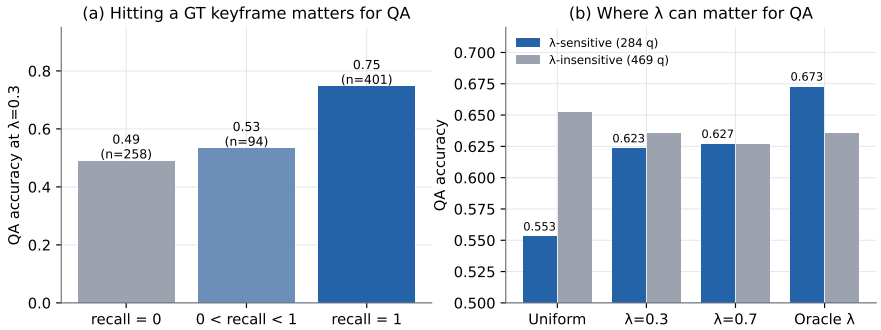
* paired, on its own test questions; 95% CI $[-1.5, +2.3]$, not significant

- The learned weight and the default give **the same accuracy**.
- Even the oracle weight adds only **1.8 points**.
- The best-recall weight (0.7) answers slightly **worse** than 0.3.

So

Better recall@8 does not automatically mean better answers.

Q4: it matters for some questions



(a) Hitting the answer moment matters: accuracy **0.49** with no hit vs. **0.75** with all hits.
(b) On the 284 questions where λ changes the frames, the oracle weight answers **5 points** more; elsewhere nothing changes.

Four mistakes we fixed

Each one ran without errors but gave wrong results.

Problem	What happened
Frames vs. seconds	Annotations are frame numbers, scores are per second: 94% of questions got zero recall.
Annotation frame rate	LongVideoBench counts frames at a fixed **24 fps**, not the real 30 fps; annotations were placed too early.
Short videos dropped	Videos $\leq$16 s could not fill VSI's 4$\times$4 grid and were silently skipped (197 questions).
Train vs. test weight	Trained on the average λ but tested on the most likely one, which picked $\lambda = 0$ or 1: learned weights looked **3–5 points worse** than the default.

Summary

1. **The best weight depends on the question type.** Text questions want more subtitle, visual questions more picture.
2. **Continual learning beats the default, but only ties a tuned constant.** The predictor collapses to one weight for all types; forgetting is small.
3. **Better frames help answers only for some questions.** About 38% of questions react to λ; there the best weight adds 5 points.
4. **Next step:** predict *per-question* differences from sparse signal, e.g., give the predictor the question type, or start from one λ per type and learn corrections.

Thank You

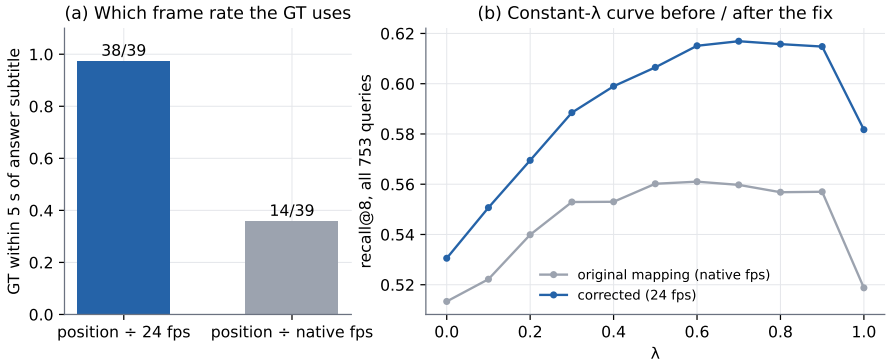
Questions?

Jianxiang He

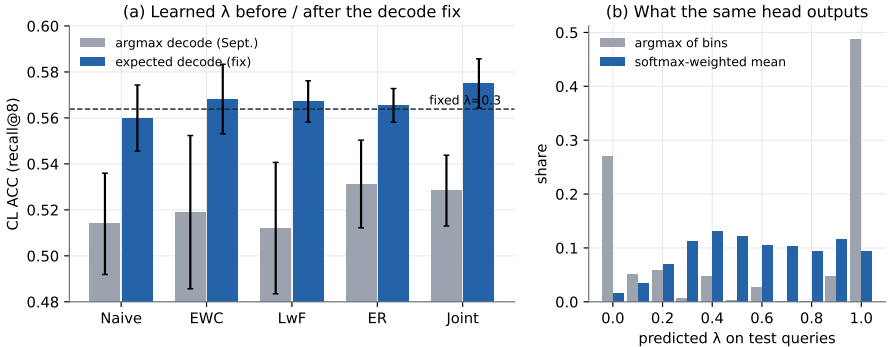
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Backup: annotation frame rate



(a) 38 of 39 annotations match the answer subtitle at 24 fps, only 14 at the real frame rate.
(b) Recall vs. λ before and after.



(a) Continual learning accuracy before and after the fix (old annotations). (b) The same predictor puts its probability on $\lambda = 0$ and $\lambda = 1$.

Backup: predictor inputs

